

AGI.Eth × AGI Jobs

ANCS Season-One — Provable Alpha Node Compute → On-Chain Reputation → Market Routing

Identity → Proof → Settlement → Governance

A public, verifiable compute tournament that turns deterministic work into settlement-grade evidence — and routes real jobs to provable competence.

The AGI.Eth Institutional Thesis

Identity → Evidence → Settlement → Governance

Identity

Roles bound to keys (ENS/DID).
Accountability by default.
Names classify actors unambiguously.

Evidence

Deterministic runs emit signed proof bundles.
Replay beats rhetoric.
Audit = replay.

Settlement

Escrow releases only after validation.
Receipts are settlement-grade artifacts.
No payout without proof.

Governance

Policy gates, upgrades, emergency brakes.
Operator override is final.
Autonomy is bounded.

ANCS — Alpha Node Compute Standard

Public Compute Olympics with cryptographic receipts

What ANCS standardizes

- Canonical benchmarks (fixed spec + fixed evaluation)
- Proof bundles (env + artifacts + hashes)
- Validator attestations (signed, replayable)
- On-chain reputation (Q_i quality, R_i reliability)
- Leaderboards & routing that anyone can recompute

Irreversibility condition

- Benchmark registered on-chain
- Runs attested by recognized validators
- Leaderboard derived from events + state only
- Router consumes Q_i/R_i directly from chain

Canonical Benchmark: AGI Jobs Green Path

Day-One Utility — deterministic operator benchmark + evidence artifacts

Operator runbook (canonical)

```
make operator:green
```

Pass criteria

- Terminal banner starts with “ Day-One Utility”
- Meets default uplift guardrail (+5.00% threshold)
- Artifacts emitted in fixed formats (JSON/HTML/PNG)

Why this is the baseline

It is already reproducible, artifact-producing, and operator-safe — the ideal “Season-Zero” anchor for public verification.

Evidence artifacts (fixed)

```
demo/AGIJobs-Day-One-Utility-Benchmark/out/...  
Benchmark/out/...
```

- report.json — machine-readable metrics
- report.html — human dashboard
- snapshot.png — visual proof
- owner_controls.json — sovereign safety state
- env.json — pinned environment digest

Proof Bundle Standard

Deterministic work → shareable bundle → hash → attestation

Bundle contents (ANCS v0.x)

- env.json — machine-checkable environment
- report.json — metrics
- report.html — dashboard
- snapshot.png — visual proof
- owner_controls.json — safety state
- score.json — normalized scoreE6
- bundle.json — manifest of hashes
- bundleHash — sha256(bundle.json)
- artifactHash — root commitment for IPFS
- nodeSig — node co-signature

Only hashes go on-chain. Heavy artifacts stay off-chain — but are cryptographically committed, replayable, and dispute-resolvable.

Validator Attestation

Recognized validators sign what they can replay

validator.alpha.club.agi.eth attests:

- benchmarkId — identity of the ruleset (manifest hash)
- runId — unique run identity
- artifactHash — commitment to the proof bundle
- envDigest + container digest — reproducibility anchor
- scoreE6 + verdict — normalized outcome
- node ENS + node signature — non-repudiation

Season-One rule

Official = quorum attestation

- 2-of-M validators must match the same tuple
- Disagreements trigger additional attestations
- Equivocation is slashable
- Finality updates reputation state

This makes lying expensive and truth cheap.

On-Chain Reputation Primitive

Reputation isn't vibes — it is attested performance

Q_i — Quality

`qualityE6(benchmarkId, node)`
= best official scoreE6
for that capability class.

R_i — Reliability

`reliabilityE6(benchmarkId, node)`
= $\lfloor 1e6 \times \text{passed} / \text{committed} \rfloor$
(missing reveals count as failures).

Why this matters

- Creates a credible supply curve for compute (who can do what, reliably)
- Lets markets route work to provable competence
- Enables dispute resolution by replay (not trust)

Routing: the Hamiltonian of Work

Select nodes by minimizing risk-adjusted “energy”

Mandatory field on every job

benchmarkId // capability class (content-addressed)

Router uses on-chain state

- $Q_i = \text{qualityE6}(\text{benchmarkId}, \text{node})$
- $R_i = \text{reliabilityE6}(\text{benchmarkId}, \text{node})$
- Optionally gate by minimum thresholds

$$H_i = \alpha \cdot \text{cost} + \beta \cdot \text{latency} - \gamma \cdot Q_i - \delta \cdot R_i + \varepsilon \cdot \text{risk}$$

Job flow

- Post job with benchmarkId
- Router selects node by Q_i/R_i
- Node executes and returns proof
- Validator verifies and settles
- Reputation updates — feedback loop

Fastest Path: JobTagger Sidecar

Immediate composition without replacing AGIJobManager

1) Create Job

Existing AGIJobManager
(unchanged contract).

2) Tag benchmarkId

JobTagger emits JobTagged
(jobId → benchmarkId).

3) Route + Settle

Router reads Q_i/R_i
then selects node; settle as usual.

Why this is minimal risk

- No migration pressure: old jobs continue uninterrupted
- benchmarkId becomes on-chain join key immediately
- Later: “one signature” batching via smart accounts

Once stable, deploy AGIJobManager vNext with benchmarkId native — migrate voluntarily.

Season-One Hardening

Make cheating a dominated strategy

Quorum

Official score requires
2-of-M matching validator attestations.
Disagreement → additional attestations.

Stake + Slashing

Validators post stake.
Equivocation or wrong-side attestation is slashable.
Withdraw delay enforces accountability.

Commit-Reveal

Nodes commit to N runs.
Missing reveals count as failures.
Reliability reflects reality, not cherry-picking.

Dominance condition

$$E[\text{cheat}] = F - p \cdot S \Rightarrow \text{choose parameters so } p \cdot S > F$$

Truth becomes the profitable equilibrium.

ENS Role Lattice

Names classify actors; keys enforce accountability

Sovereigns

*.alpha.agi.eth · *.agi.eth

- Policy gates & upgrades
- Emergency brakes (pause/reset)
- Canonical registries

Validators

*.alpha.club.agi.eth · *.club.agi.eth

- Replay deterministically
- Attest (quorum)

Agents

*.alpha.agent.agi.eth · *.agent.agi.eth

- Create / claim jobs
- Orchestrate nodes
- Produce settlement proofs

Nodes

*.alpha.node.agi.eth · *.node.agi.eth

- Execute deterministically
- Emit proof bundles

Autopoiesis: the Self-Maintaining Node

A sovereign digital worker that preserves its operational identity

Metabolic loop

- Compute → Work → Proof → Settlement → Reputation
- Heartbeat benchmark to detect drift
- Pinned environments for replayability
- Signed telemetry for accountability

Immune system

- Quorum detects outliers
- Slashing corrects misbehavior
- Governance applies brakes
- Replay resolves disputes

Stability scales when maintenance is automated.

Launch Sequence

Velocity first, then harden

Season-Zero deliverables

- Deploy JobTagger sidecar (benchmarkId join key)
- Deploy benchmark registry + attestation contract
- Publish public leaderboard from RunAttested events

Season-One upgrades

- 2-of-M quorum finalization (official runs only)
- Validator staking + slashing (equivocation / wrong-side)
- Commit-reveal seasons for nodes (anti cherry-picking)
- Optional on-chain gating: minQualityE6 / minReliabilityE6

Kardashev-Scale Coordination

Energy → Work → Proof → Trust

The civilization engine

When compute becomes measurable, replayable, and settleable, coordination scales like physics: a stable macro-order emerges from local rules.

- Hamiltonian routing shapes global behavior via local incentives
- Gibbs sampling enables exploration without chaos
- Slashing is an energy barrier against adversarial states
- Governance is the emergency brake: bounded autonomy

ANCS is the bridge from demos to economies.

Enter the Tournament

From proof to profit — the shortest path

Node loop

- Run the benchmark (Green Path / Day-One Utility)
- Publish the proof bundle (IPFS or equivalent)
- Request validator attestation (quorum)
- Gain on-chain Q_i/R_i for benchmarkId
- Receive routed AGI Jobs with benchmarkId

```
make operator:green # canonical run
```

Next: one-signature onboarding (smart account batching)

The promise

A market where jobs route to provable competence — and competence is measured by replay, not reputation games.

Identity → Proof → Settlement → Governance

Now make it public.